

PHISHING URL SECURITY AUDIT REPORT

Executive Cybersecurity Risk Assessment & Explainable AI (XAI) Diagnostic Audit

SECURITY AUDIT
REF: PG-AUDIT-00353

1. REPORT INFORMATION

Report Title:	Phishing URL Security Audit Report	Scan ID:	#353 (PG-SCAN-00353)
Date and Time:	August 20, 2026 at 15:43:24 UTC	Domain:	paypal-security-update.account-verify.xyz
Analyzed URL:	http://paypal-security-update.account-verify.xyz/signin.php	Model Engine:	Random Forest (XAI Calibrated)

2. FINAL VERDICT

■ CRITICAL PHISHING THREAT DETECTED RISK LEVEL: CRITICAL

This URL exhibits verified malicious deception patterns characteristic of active credential harvesting or brand impersonation.

PREDICTION PHISHING	RISK LEVEL CRITICAL	CONFIDENCE SCORE 85.0%	PHISHING PROB. 85.0%	LEGITIMATE PROB. 15.0%
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3. URL SECURITY ANALYSIS

Security Characteristic	Diagnostic Status	Security Characteristic	Diagnostic Status
HTTPS	No (Insecure HTTP)	Domain	paypal-security-update.account-verify.xyz
URL Length	59 characters (Normal)	Domain Length	41 characters
Number of Subdomains	1 level(s)	Suspicious Keywords	Yes (3 keywords: signin , verify , account)
IP Address Usage	No (DNS Host)	Suspicious URL Structure	Yes (Hyphenated Brand Spoofing)
TLD Abuse Rating	80% Risk (High-Risk Abuse TLD)	Shannon Entropy	4.35 bits (High Randomness)

4. WHY THIS RESULT WAS GIVEN

Main Factors Influencing Decision (5 Detected):

- **Use of HTTP / Insecure Transport:** URL uses unencrypted plaintext HTTP protocol instead of encrypted HTTPS.
- **Suspicious Keywords Detected:** Found 3 sensitive credential/security keyword(s) ('signin', 'verify', 'account') in the URL.
- **Hyphenated Brand Spoofing:** Domain structure contains deceptive brand combinations (combosquatting).
- **High-Risk Top-Level Domain:** Registered under a TLD frequently associated with spam and automated phishing campaigns.
- **Unusual URL Characteristics (Entropy 4.35):** Elevated randomness indicates algorithmic domain generation.

5. EXPLAINABLE AI SUMMARY

Concise mathematical summary of the top 3–5 features from Explainable AI (SHAP & LIME) influencing the model's decision:

Rank	Influential Feature	Observed Value	Risk Impact Direction	Short Impact Explanation
#1	TLD Risk Factor	0.8	+Risk (Pushed Phishing)	Registered under a high-risk Top-Level Domain frequently associated with spam and phishing.
#2	Subdomain Count	1	-Risk (Pushed Safe)	Clean, standard top-level domain hierarchy without excessive nesting.
#3	Path Length	11	-Risk (Pushed Safe)	Length of the directory and query path.
#4	Suspicious Keywords Count	3	+Risk (Pushed Phishing)	Contains 3 deceptive security/credential harvest keywords.
#5	HTTPS Protocol	0	+Risk (Pushed Phishing)	Insecure HTTP protocol strongly elevates the risk of spoofing.

6. SECURITY RECOMMENDATION

■ CRITICAL SECURITY DIRECTIVES (HIGH RISK / PHISHING DETECTED)

• Do not enter username or password:

This URL exhibits credential harvesting patterns.

• Do not provide personal or financial information:

Never enter credit card, banking, or identity data.

• Do not download files from the URL:

Files hosted on this domain may contain malware or spyware.

• Verify the domain using a trusted source:

Navigate to official services directly via known bookmarks or official search.

• Close or report the suspicious page if appropriate:

Terminate active browser tabs and report this link to your security team.

7. SHORT SECURITY SUMMARY

URL:	http://paypal-security-update.account-verify.xyz/signin.php	Risk:	CRITICAL
Verdict:	PHISHING	Confidence:	85.0% (Phishing: 85.0%)
Main Reason:	Detected 5 risk factors: Insecure HTTP, Credential keywords, Malicious pattern matching (Phishing Probability: 85.0%).	Recommended Action:	DO NOT PROCEED. Do not enter passwords or submit sensitive details.